CT-1 (PH3101)

~~1 (a)~~

1(a) $L = \frac{1}{2}m(\dot{r}^2 + \omega^2 r^2) - \frac{1}{2}\frac{mk}{a}(r-a)^2$

$\therefore \frac{\partial L}{\partial r} = \frac{1}{2}m \cdot 2\omega^2 r - \frac{mk}{a}(r-a)$

$= m\omega^2 r - \frac{mk}{a}r + mk$

$\therefore \frac{\partial L}{\partial \dot{r}} = m\dot{r}$

$\therefore \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) = m\ddot{r}$

$\therefore \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) - \frac{\partial L}{\partial r} = 0.$

$\Rightarrow m\ddot{r} - m\omega^2 r + \frac{mk}{a}r - mk = 0$ — Langrange EOM

(b) ~~(X)~~

$E = T + V.$

$= \frac{1}{2}m(\dot{r}^2 + \omega^2 r^2) + \frac{1}{2}\frac{mk}{a}(r-a)^2$

$\therefore \frac{dE}{dt} = m\dot{r}\ddot{r} + m\omega^2 r\dot{r} + \frac{mk}{a}(r-a)\dot{r}$

$= \dot{r}\left[m\ddot{r} + m\omega^2 r + \frac{mkr}{a} - mk\right]$

(c) $E = \dot{r}\frac{\partial L}{\partial \dot{r}} - L$ $\qquad E = \frac{1}{2}m(\dot{r}^2 - \omega^2 r^2) + \frac{1}{2}\frac{mk}{a}(r-a)^2$

$= \dot{r}m\dot{r} - \frac{1}{2}m(\dot{r}^2 + \omega^2 r^2) + \frac{1}{2}\frac{mk}{a}(r-a)^2$

$\therefore \frac{dE}{dt} = 2m\dot{r}\ddot{r} - \dot{r}\left[m\ddot{r} + m\omega^2 r - \frac{mkr}{a} + mk\right]$. (from cancelled calculat^n).

$= \dot{r}\left[m\ddot{r} - m\omega^2 r + \frac{mkr}{a} - mk\right] = 0$

(from Langrange EOM)

Hence, total E is conserved

(d) $\ddot{r} + r\left(\frac{k}{a} - \omega^2\right) = k$

~~Let r =~~ CF: $\left[\ddot{r} + r\left(\frac{k}{a} - \omega^2\right) = 0\right]$

Let $r = A e^{\alpha t}$

$\therefore \ddot{r} = \alpha^2 e^{\alpha t}$

$\therefore \alpha^2 + \left(\frac{k}{a} - \omega^2\right) = 0$

$\Rightarrow \alpha = \pm\sqrt{\omega^2 - \frac{k}{a}}$

$\therefore$ ~~$\boxed{r(t) = e^{\alpha t}}$ — CF~~ $\therefore \boxed{r(t) = A_1 e^{\alpha t} + A_2 e^{-\alpha t}}$ — CF

PI: $\left[\ddot{r} + r\left(\frac{k}{a} - \omega^2\right) = k\right]$

Let $r = \dfrac{k}{\frac{k}{a} - \omega^2}$ $\Rightarrow \ddot{r} = 0.$

— PI

$\therefore$

$\therefore r(t) = CF + PI$

$$\boxed{r(t) = A_1 e^{\alpha t} + A_2 e^{-\alpha t} + \frac{k}{\frac{k}{a} - \omega^2}} \quad ; \quad \alpha = \sqrt{\omega^2 - \frac{k}{a}}$$

(Ans)

(e) Total Energy (E) of the system is conserved.

③ (1) $v = 0$ (free particles)

$x = r\cos\theta \quad ; \quad y = r\sin\theta$

$\dot{x} = -r\sin\theta\,\dot{\theta} + \cos\theta\,\dot{r}$

$\dot{y} = r\cos\theta\,\dot{\theta} + \sin\theta\,\dot{r}$

$\therefore \; v^2 = \dot{x}^2 + \dot{y}^2$

$= (-r\sin\theta\,\dot{\theta} + \dot{r}\cos\theta)^2 + (r\cos\theta\,\dot{\theta} + \dot{r}\sin\theta)^2$

$v^2 = r^2\dot{\theta}^2 + \dot{r}^2$

$\therefore \; L = T - V = \frac{1}{2} m (r^2\dot{\theta}^2 + \dot{r}^2)$

$\therefore$ For r:

$\frac{\partial L}{\partial r} = mr\dot{\theta}^2 \; ; \; \frac{\partial L}{\partial \dot{r}} = m\dot{r} \; ; \; \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) = m\ddot{r}$

$\therefore \; \boxed{m\ddot{r} - mr\dot{\theta}^2 = 0}$ — (I)

For θ:

$\frac{\partial L}{\partial \theta} = 0 \; ; \; \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}$

$\therefore \; \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = 0 \Rightarrow \boxed{mr^2\dot{\theta} = \text{constant}}$ — (II)

(I) & (II) are the EOM.

(II) $mr^2\dot{\theta} = L \Rightarrow \boxed{\dot{\theta} = \frac{L}{mr^2}}$

(ii) If the particle moves in a straight line:

$$y = mx + c.$$

$$\Rightarrow \quad r\sin\theta = mr\cos\theta + c$$

$$\Rightarrow \quad \dot{r}\sin\theta + r\cos\theta\,\dot{\theta} = m[\dot{r}\cos\theta - r\sin\theta\,\dot{\theta}]$$

$$\Rightarrow \quad \ddot{r}\sin\theta + \dot{r}\cos\theta\,\dot{\theta} + \dot{r}\cos\theta\,\dot{\theta} - r\sin\theta\,\dot{\theta}^2 + r\cos\theta\,\ddot{\theta}$$

$$= m[\underbrace{\ddot{r}\cos\theta - \dot{r}\sin\theta\,\dot{\theta} - \dot{r}\sin\theta\,\dot{\theta} - r\cos\theta\,\dot{\theta}^2}_{(a)} - r\sin\theta\,\ddot{\theta}]$$

$$\Rightarrow \quad \sin\theta\,[\underbrace{(\ddot{r} - r\dot{\theta}^2)}_{(a)} + m\underbrace{(2\dot{r}\dot{\theta} + r\ddot{\theta})}_{(b)}]$$

$$+ \cos[\underbrace{(2\dot{r}\dot{\theta} + r\ddot{\theta})}_{(b)} - m\underbrace{(\ddot{r} - r\dot{\theta}^2)}_{(a)}] = 0$$

$\sin\theta$ & $\cos\theta$ are linearly independent. Hence if

$$a\sin\theta + b\cos\theta = 0 \Rightarrow (a, b) = (0, 0)$$

Now, $mr^2\dot{\theta} = \text{constant}$.

$$\Rightarrow \quad r^2\dot{\theta} = \text{constant}.$$

$$\Rightarrow \quad 2r\dot{r}\dot{\theta} + r^2\ddot{\theta} = 0.$$

$$\Rightarrow \quad (2\dot{r}\dot{\theta} + r\ddot{\theta})r = 0.$$

This is true for all r. Hence $\boxed{2\dot{r}\dot{\theta} + r\ddot{\theta} = 0}$ — (iii)

Using (i) & (iii); the coefficients of $\sin\theta$ & $\cos\theta$ (combinations of (a) & (b)) are 0.

Hence, the equation $y = mx + c$ is satisfied & the particle moves in a straight line

② (i) $L = \frac{1}{2} I_1 (\dot{\theta}^2 + \dot{\phi}^2 \sin^2\theta) + \frac{1}{2} I_3 (\dot{\psi} + \dot{\phi}\cos\theta)^2 - mgl\cos\theta$

$\therefore \frac{\partial L}{\partial \phi} = 0 \quad ; \quad \frac{\partial L}{\partial \psi} = 0.$

$\therefore \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\phi}}\right) = 0 \Rightarrow \frac{\partial L}{\partial \dot{\phi}} = \text{constant}$ — (i)

$\therefore \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\psi}}\right) = 0 \Rightarrow \frac{\partial L}{\partial \dot{\psi}} = \text{constant}$ — (ii)

$\therefore \frac{\partial L}{\partial \dot{\phi}} = I_1 \sin^2\theta\, \dot{\phi} + I_3 (\dot{\psi} + \dot{\phi}\cos\theta)\cos\theta$ — (iii)

$\therefore \frac{\partial L}{\partial \dot{\psi}} = I_3 (\dot{\psi} + \dot{\phi}\cos\theta)$ — (iv)

(iii) & (iv) are the constants of motion.

(ii) $\therefore \frac{\partial L}{\partial \theta} = I_1 \sin\theta\cos\theta\, \dot{\phi}^2 + I_3 (\dot{\psi} + \dot{\phi}\cos\theta)(-\sin\theta\, \dot{\phi}) + mgl\sin\theta$

$\therefore \frac{\partial L}{\partial \dot{\theta}} = I_1 \dot{\theta}$

$\therefore \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0.$

$\Rightarrow I_1 \ddot{\theta} = I_1 \sin\theta\cos\theta\, \dot{\phi}^2 + I_3 (\dot{\psi} + \dot{\phi}\cos\theta)(-\sin\theta\, \dot{\phi}) + mgl\sin\theta$

Now, $I_3 (\dot{\psi} + \dot{\phi}\cos\theta) = p_\psi$.

$I_1 \sin^2\theta\, \dot{\phi} + p_\psi \cos\theta = p_\phi$

$$\Rightarrow \boxed{\dot{\phi} = \frac{p_\phi - p_\psi \cos\theta}{I_1 \sin^2\theta}}$$

$$\therefore I_1 \ddot{\theta} = I_1 \sin\theta \cos\theta \left[\frac{p_\phi - p_\psi \cos\theta}{I_1 \sin^2\theta} \right]^2 + p_\psi \sin\theta \left[\frac{p_\phi - p_\psi \cos\theta}{I_1 \sin^2\theta} \right] + mgl \sin\theta$$ (Ans)

(iii) The system is integrable

There are 3 dof - θ, ϕ, ψ.

There are 3 ~~const~~ independent constants of motion.

(a) Total energy - as L is independent of time

(b) ~~p_ϕ~~ p_ϕ ; (c) p_ψ.